

Data Quality Assessment Log

Project Title: Customer Analytics Platform for Strategic Customer Retention and Revenue Growth

Dataset: Olist Brazilian E-commerce Dataset

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Status definitions

- **PASS** - No data-quality issue identified.
- **REVIEW** - Result is not necessarily an error but requires business interpretation.
- **FAIL** - Data violates an expected integrity/domain/business rule and requires remediation.

ID	Finding	Result	Status	Interpretation
DQ-001	Duplicate customer IDs	0	PASS	Customer table has unique customer_id
DQ-002/003	Multiple customer IDs per unique customer	2,997	REVIEW	Structural distinction between customer account and underlying customer
DQ-004	Missing customer identifiers	0	PASS	Complete
DQ-005	Duplicate order IDs	0	PASS	Order grain confirmed
DQ-006	Missing order/customer IDs	0	PASS	Complete
DQ-007	Orphan orders	0	PASS	Referential integrity confirmed
DQ-008	Duplicate order-item keys	0	PASS	Transaction grain confirmed
DQ-009	Missing order-item identifiers	0	PASS	Complete
DQ-010	Orphan order items → orders	0	PASS	Referential integrity confirmed
DQ-011	Orphan products	0	PASS	Referential integrity confirmed
DQ-012	Orphan sellers	0	PASS	Referential integrity confirmed
DQ-013	Invalid item prices	0	PASS	No negative/null prices
DQ-014	Invalid freight values	0	PASS	No negative/null freight
DQ-015	Missing payment identifiers	0	PASS	Complete
DQ-016	Orphan payments	0	PASS	Referential integrity confirmed
DQ-017	Invalid payment values	0	PASS	No negative/zero/missing values
DQ-018	Payment type domain	5 valid categories	PASS	No unexpected payment types
DQ-019	Invalid payment installments	2	REVIEW	Two credit-card records have 0 installments

ID	Finding	Result	Status	Interpretation
DQ-020	Invalid review scores	0	PASS	Scores conform to 1–5 domain
DQ-021	Missing review identifiers	0	PASS	Complete
DQ-022	Orphan reviews	0	PASS	Referential integrity confirmed
DQ-023	Duplicate review IDs	789	REVIEW	Review ID is not unique; requires understanding of review grain
DQ-024	Approval before purchase	0	PASS	Temporal consistency
DQ-025	Carrier before approval	1,359	INVESTIGATE	Potential timestamp sequencing issue
DQ-026	Customer delivery before carrier delivery	23	REVIEW	Small number of temporal anomalies
DQ-027	Estimated delivery before purchase	0	PASS	Temporal consistency
DQ-028	Delivery before purchase	0	PASS	Temporal consistency
DQ-029	Delivered without customer delivery date	8	REVIEW	Business-rule inconsistency
DQ-030	Delivered without carrier date	2	REVIEW	Business-rule inconsistency
DQ-031	Canceled with delivery date	6	REVIEW	Possible status/data inconsistency
DQ-032	Unavailable with delivery date	0	PASS	Consistent
DQ-033	Order status domain	8 expected statuses	PASS	No unexpected statuses
DQ-034	Duplicate product IDs	0	PASS	Product grain confirmed
DQ-035	Missing product IDs	0	PASS	Complete
DQ-036	Zero/negative/missing weights	4 zero, 2 missing	REVIEW	Small number of anomalous/missing physical attributes
DQ-037	Invalid dimensions	0	PASS	Valid
DQ-038	Missing product categories	610	REVIEW	Category information incomplete
DQ-039	Duplicate seller IDs	0	PASS	Seller grain confirmed
DQ-040	Missing seller IDs	0	PASS	Complete
DQ-041	Missing seller geography	0	PASS	Complete
DQ-042	Missing geolocation fields	0	PASS	Complete
DQ-043	Invalid latitude	0	PASS	Valid geographic range
DQ-044	Invalid longitude	0	PASS	Valid geographic range

ID	Finding	Result	Status	Interpretation
DQ-045	ZIP → multiple states	8 ZIP prefixes	REVIEW	Geographic mapping ambiguity
DQ-046	Duplicate category translations	0	PASS	Reference table consistent
DQ-047	Missing category translations	0	PASS	Translation table complete
DQ-048	Untranslated product categories	13	REVIEW	Small number of category mapping gaps
DQ-049	Orders without items	775	REVIEW	Requires investigation by order status
DQ-050	Orders without payment	1	REVIEW	Isolated missing payment record
DQ-051	Orders without review	768	REVIEW	Not necessarily a quality issue; reviews are optional
DQ-052	Orphan sellers	0	PASS	Referential integrity confirmed
DQ-053	Orphan products	0	PASS	Referential integrity confirmed

Data Quality Issues Requiring Action

Issue	Source	Finding	Business Impact	Silver-Layer Treatment
Multiple customer_id per customer_unique_id	Customers	2,997 underlying customers are associated with multiple customer_id values	Using customer_id as the behavioral customer key could overstate customer counts and distort repeat-purchase metrics	Retain both identifiers. Use customer_unique_id as the customer-level behavioral key
Zero payment installments	Payments	2 credit-card records have 0 installments	Could affect installment-based payment analysis	Retain the records; flag as anomalous. Do not alter without evidence that they are erroneous
Non-unique review_id	Reviews	789 review IDs occur more than once; repeated IDs correspond to different orders	Treating review_id as a primary key would incorrectly collapse legitimate review records	Define review grain using review_id + order_id rather than review_id alone
Carrier timestamp precedes approval timestamp	Orders	1,359 records fail the simple	Naive delivery-cycle calculations could produce	Do not overwrite timestamps. Investigate/flag the

Issue	Source	Finding	Business Impact	Silver-Layer Treatment
		carrier_date >= approval_date test	misleading processing times	sequencing anomaly and use robust delivery metrics based on validated timestamps
Customer delivery precedes carrier delivery	Orders	23 records	Can produce negative carrier-to-customer delivery duration	Flag anomalous records; exclude from carrier-to-customer duration calculations unless validated
Delivered orders missing customer delivery date	Orders	8 records	Delivery-time metrics cannot be calculated for these orders	Preserve NULL; exclude from delivery-duration calculations
Delivered orders missing carrier date	Orders	2 records	Carrier transit-time metrics cannot be calculated	Preserve NULL; exclude from carrier-duration calculations
Canceled orders with delivery date	Orders	6 records	Potential status/timestamp inconsistency	Preserve raw status/timestamps; flag for operational analysis rather than changing status
Zero/missing product weight	Products	4 zero-weight and 2 missing-weight products	Can distort weight-based logistics analysis	Convert confirmed invalid zero values to NULL/flag after validation; retain missingness
Missing product categories	Products	610 products (1.85%) have no category	Affects category-level product and customer analysis	Resolve where possible; otherwise assign controlled Unknown category in Silver
ZIP prefix mapped to multiple states	Geolocation	8 ZIP prefixes have multiple state mappings	Direct geographic joins could produce ambiguous or duplicated records	Build a deterministic ZIP-prefix geographic mapping before joining
Untranslated product categories	Products / Translation	13 product category values	Small number of products may remain unavailable	Map where possible; assign controlled

Issue	Source	Finding	Business Impact	Silver-Layer Treatment
		have no English translation	for English-category analysis	Unknown/Untranslated label where unresolved
Orders without order items	Orders	775 orders; 603 unavailable, 164 canceled, 5 created, 2 invoiced, 1 shipped	Treating every order as a complete transaction could distort item/revenue analysis	Exclude orders without items from item-based revenue calculations; retain them for order-status analysis. Investigate the single shipped order separately
Order without payment	Orders / Payments	1 order has no payment record	Could cause undercounting if payment table is used as the sole revenue source	Investigate the order; preserve the order and avoid assuming missing payment means zero revenue
Orders without reviews	Orders / Reviews	768 orders lack reviews	Review coverage is incomplete, but reviews are optional	No imputation. Treat review metrics as applicable only to orders/customers with reviews

DQA Summary

The Data Quality Assessment found that the Olist dataset is structurally robust, with primary transactional identifiers generally unique, complete referential relationships, valid domain values, and extensive geographic coverage. The principal issues identified were not widespread corruption but a combination of structural characteristics, limited missingness, and a small number of temporal and business-rule anomalies.

The most important structural finding is the distinction between `customer_id` and `customer_unique_id`, which requires `customer_unique_id` to be used for customer-level behavioral analysis. Similarly, repeated `review_id` values indicate that `review_id` alone should not be treated as a unique review key.

Several timestamp inconsistencies were identified, particularly carrier timestamps preceding approval timestamps. Investigation indicates that these should be treated as source-data anomalies rather than automatically corrected. Delivery metrics will therefore be constructed using validated timestamps and appropriate exclusion/flagging rules.

Overall, the dataset is suitable for downstream analytical use after applying the documented transformation and validation rules. The findings from the DQA will directly determine the Silver-layer data model and transformation logic.